

Algebra II with Trigonometry

Chapter 5.3

Mental Math

Saved tutoring response with MathJax rendering. Original PDF file:
[4401269_alg-2trig-h-chp-5-3-note-docx.pdf](#)

Here are 10 mental math questions based on the **core trigonometric concepts** from your PDF. These can be answered without paper and pencil if you know the key facts and relationships!

1. Quadrants & Signs

If $\sin \theta$ is positive and $\cos \theta$ is negative, in which quadrant does θ lie?

2. Reciprocal Identities

If $\sin \theta = 1/2$, what is $\csc \theta$?

3. Reference Angles

What is the reference angle for 150° ?

4. Pythagorean Identity

If $\sin^2 \theta = 1/4$, what is $\cos^2 \theta$?

5. Quadrantal Angles

At what angles between 0° and 360° is $\tan \theta$ undefined?

6. Relationship Between Functions

If $\tan \theta = 2$, what is $\cot \theta$?

7. Positive Functions by Quadrant

In which quadrant(s) is $\sec \theta$ positive?

8. Negative Functions by Quadrant

In which quadrant is $\tan \theta$ negative while $\cos \theta$ is positive?

9. Simple Area of Triangle (Formula Recall)

If a triangle has sides 5 and 12 with an included angle of 90° , what is the area?

10. Angle Coterminal with Axes

Name an angle between 0° and 360° that is coterminal with the y-axis.

Would you like the answers and explanations as well?